

Functional insights of antibiotic resistance mechanism in *Helicobacter pylori*: Driven by gene interaction network and centrality-based nodes essentiality analysis

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ARTICLE INFO

Keywords:

Gene interaction network

Helicobacter pylori

Antibiotic resistance

Gene enrichment analysis

ABSTRACT

Increased antibiotic resistance in *Helicobacter pylori* (*H. pylori*), a major human pathogen, constitutes a substantial threat to human health. Understanding the pathophysiology and development of antibiotic resistance can aid our battle with the infections caused by *H. pylori*. The aim of this study is to discover the high-impact key regulatory mechanisms and genes involved in antimicrobial drug resistance (AMR). In this study, we constructed a functional gene interaction network by integrating multiple sources of data related to antibiotic resistant genes (number-77) from *H. pylori*. The gene interaction network was assortative, with a hierarchical, scale-free topology enriched in a variety of gene ontology (GO) categories and KEGG pathways. Using an iterative clustering methodology, we identified a number of communities in the AMR gene network that comprised nine genes (*sodB*, *groEL*, *gyrA*, *recA*, *polA*, *tuf*, *infB*, *rpsJ*, and *gyrB*) that were present at the deepest level and hence were key regulators of AMR. Further, an antibiotic-resistant gene network-based centrality analysis revealed superoxide dismutase (*sodB*) as a bottleneck node in the network. Our findings suggested that *sodB* is critically enriched in the cellular response to oxidative stress, removal of superoxide radicals, cellular oxidant detoxification processes, cellular component biogenesis, response to reactive oxygen species, urea metabolic process, nitrogen cycle metabolic process and reactive oxygen species metabolic process. We demonstrated how the *sodB*, which are involved in the response to reactive oxygen species, urea metabolic process, nitrogen cycle metabolic process, reactive oxygen species metabolic process, regulated by Fur gene/proteins, claim a major authority over regulation and signal propagation in the AMR.

1. Introduction

Helicobacter pylori (*H. pylori*), a Gram negative and transmissible bacterium due to its ability to tolerate the acidic gastric environment remains as the most effective human gastric pathogen, colonising the stomach in over 50% of world's population [1]. The WHO has issued a list of microorganisms in need of new antibiotics. Clarithromycin-resistant *H. pylori* was placed in the high priority category, which consists of 12 families divided into three categories: critical, high, and medium [2]. Antibiotics have been used indiscriminately for decades, resulting in the development of multidrug resistance (MDR). Antibiotic resistance concerns worldwide health regulatory

organisations, and researchers are working round the clock to solve the issue. The World Health Organization (WHO) has identified a number of species that deserve differing degrees of antibiotic resistance research attention [1].

The fundamental molecular foundation of drug resistance in *H. pylori* is specific single nucleotide polymorphisms, and the pressure that dictates the selection of resistant strains is connected to frequent and/or inappropriate antibiotic use [3]. Antibiotic resistance in *H. pylori* species (with the possibility of resistance mutation) is caused by a variety of biological factors, the most common of which are structural changes in genetic sequences that disrupt antibiotic cellular activity by either altering the drug target or inhibiting drug activation within the cells.

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<https://doi.org/10.1016/j.micpath.2022.105737>

Received 12 July 2022; Received in revised form 5 August 2022; Accepted 19 August 2022

Available online 28 August 2022

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Drug resistance in *H. pylori* appears to be an emerging field that requires more investigation. The rapid advancement of next-generation sequencing technologies, which enable holistic and multi-omics methods, is poised to advance existing understanding and management capabilities of *H. pylori* resistance pathways. To combat this global problem, we need not only novel antibiotics and treatment strategies, but also a greater knowledge of how resistant bacteria originate and spread, as well as enhanced diagnostic technologies to assist clinicians in further optimising presently available regimens. The fundamental processes that drive treatment resistance in *H. pylori* have made significant progress.

Gene interaction networks are one of the computational systems biology tools for understanding AMR in pathogenic and opportunistic microorganisms, as well as other biological disorders. The collected AMR genes, processed for gene interaction network and clustering analysis to examine various topological metrics as well as the functional relatedness of AMR genes to other functional partners [4]. With the metrics of centrality, betweenness, and degree, network topological structure could reveal essential information from its nodes or edges. Hub nodes, also known as prominent nodes, play an important role in the complex biological network [5]. Exploratory network structure and comprehending topological features could lead to the discovery of new nodes in a complex biologic system, as well as fresh insights into molecular mechanism elucidation [6]. Henceforth, exploring the functional association between the AMR genes would drive our knowledge in understanding the mechanism that thrive in acquiring the resistance in pathogens. Our findings contribute to a better understanding of *H. pylori* antibiotic resistance mechanism at the molecular level.

2. Computational tools and methods

2.1. Data acquisition

The antibiotic resistant gene data was obtained various databases like Pathway-Systems Resource Investigation Centre (PATRIC) [7], Antibiotic resistance database (ARDB) [8], Comprehensive antibiotic resistance database (CARD) [9] and by literature survey using PubMed.

2.2. Interaction data and construction of gene interaction network

The AMR genes data was used to construct a gene interaction network based on the statistical prediction. The STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) (<https://string-db.org/>) database, which contains experimental evidence, is one of the most well-known roles of protein-protein interaction (PPI) databases [10]. To find and include these connections, researchers employed a combination of literature searches, database mining, high-throughput experimental data, and co-expressed gene studies. This STRING database contains interaction data for all known and expected PPI, as well as direct (physical interactions with experimental evidence) and indirect (interactions predicted using computational methodologies) PPI. STRING generates a confidence score or combination score for each interaction in the database based on the available data curation source. The levels of confidence range from 0 to 1, with higher confidence numbers suggesting more evidence. String splits the interactions into four categories based on the confidence scores: highest (above 0.90), high (0.7–0.89), medium (0.4–0.69), and low (0.1) [11,12]. By providing a list of gene as input, the STRING searches for the closest neighbours with direct relationships. Furthermore, the gene interaction network was visualised using the Cytoscape 3.9.1 tool, and network node attributes were calculated using default parameters [13].

2.3. Characterization of centrality-based network topological properties

The molecular organisation can be represented by a network of variably connected nodes. Each node represents a protein, while the

edges indicate dynamic interactions. As a result, nodes receive mathematical function input and output values [14]. Topological features aid in the comprehension of network components (nodes and edges) as well as related substructures [15]. The nodes in a network were evaluated using two key topological characteristics from network theory: connectedness degree (k) and betweenness centrality. In addition, the proximity centrality quantifies the average distance between nodes. The shortest path is defined as the length of all geodesics from or to a network vertex. The node with the highest betweenness centrality has a considerable impact on network traffic and hence plays an important role as a global attribute [16]. We were able to characterise the physical properties of complex networks by observing at the behaviour of the following topological parameters. The Network Analyzer and CytoNCA tools in Cytoscape 3.9.1 were used to determine these parameters [17].

2.3.1. Degree distribution, $p(k)$

The degree (k) of a node in a graph or network, $G = (E, V)$, is defined as the number of edges connecting to that node, where E and V are node and edge sets, respectively. If $V = \{n\}$ and $E = \{e_{ij}; i, j, i \neq j\}$, the probability of the degree distribution is as follows:

$$p(k) = \frac{n_k}{N}$$

where n_k is the number of nodes with a degree of k and N is the network's size.

2.3.2. Clustering coefficient, $c(k)$

A clustering coefficient helps measure the clustering propensity of nodes in a network and determines the quantity of connected triangles in a network (nodes in a network tend to bind their neighbours and form a number of triangular motifs. The network gets strength as a result of its internal connectivity and clustering [16]. Clustering coefficient for a node, i (with degree k_i), in an undirected graph is the ratio of total number of triangular motifs formed by it with nearest neighbours to total number of triangular motifs in the network, as follows:

$$c(k) = \frac{2m_i}{k_i(k_i - 1)}$$

where m_i is the total number of edges between a node i and its nearest neighbours. The clustering coefficient is approximately constant in scale-free networks, but in hierarchical networks, we can discern a power law versus degree $C(k) \sim k^{-a}$, with $a \sim 1$.

2.3.3. Betweenness centrality, $C_B(k)$

$C_B(k)$ is proportional to the number of shortest/geodesic paths travelling through a node, and hence represents the volume of data flowing through it. It is far more important in determining the essentiality of a gene/protein than its network degree. In a PPI network for a disorder, nodes with a high $C_B(k)$ are common prospective therapeutic targets, and so provide a high utility purpose in understanding its aetiology. It can be calculated in a graph by dividing the total number of shortest pathways by the number of shortest paths passing through that node.

$$C_B(n_i) = \sum_{i < k} g_{jk}(n_i) / g_{jk}$$

where g_{jk} denotes the number of shortest paths between j and k , and $g_{jk}(n_i)$ the number of nodes on which i is located.

2.3.4. Closeness centrality, $C_C(k)$

$C_C(k)$ determines the rate at which information is disseminated from a node to the other nodes connected to it. It is determined as the average of the shortest path length from a node to all other nodes in the network and represents a node's closeness to all other nodes in the network.

$C_C(k)$ of a node m is mathematically defined as:

$$C_c(k) = \frac{n}{\sum_j d_{mj}}$$

2.3.5. Eigenvector centrality, $C_E(k)$

$C_E(k)$ distinguishes highly linked neighbourhoods from those with low connections, emphasising the spreading power of that node in the network and minimising the likelihood of a node with high $C_E(k)$ being located solitary. A node's mean eigenvector centrality ($C_E(m)$) in a network is proportional to the total (but not the average) of its neighbours' centrality [26].

$$C_E(m) = \frac{1}{\lambda} \sum_{j=nn(m)} n_j$$

where $nn(m)$ denotes the network's nearest neighbours, λ = eigenvalue, and v_j = eigenvector. The major eigenvector of A corresponding to the maximum positive eigenvalue, i.e., $\max$, is used to denote $C_E(k)$ score.

2.3.6. Neighbourhood connectivity distribution, $C_N(k)$

The neighbourhood connectivity of a node with degree k in a network is the average connectedness with its nearest neighbours and is mathematically represented as below [18],

$$C_N(k) = \sum_q qP(q|k)$$

$P(q|k)$ is the conditional probability of a link belonging to a node with connectivity of k pointing to a node with connectivity of q . For a hierarchical network, $C_N(k)$ follows a power law in k ; $C_N(k)k$ with 0.5, and $C_N(k)$ constant for a scale-free network. Positive denotes assortativity, whereas negative shows disassortativity.

2.4. Filtering hub genes in the network using MCODE (molecular complex Detection)-Derived protein complexes

Hub genes were filtered for noise by tracing them in clusters created using rigorous parameters in MCODE, a Cytoscape 3.9.1 plugin/application that uses an automated clustering approach to extract/identify densely connected areas or protein complexes in a PPI network [19]. In a large gene interaction network, the densely connected regions were identified based on the differential weight of nodes, density of nodes local neighbourhood and traversals arising from the seed nodes. These network clusters aid us in identifying the gene groups involved in specific biological processes and pathways.

2.5. Identification of key regulators (KRs) in the core network

Using the Louvain method of modularity (Q) maximisation for community detection in the "igraph" package [20] of R, genes present in core AMR gene interaction network with degree ≥ 20 were traced down in various sub modules or clusters at each hierarchical level up to the motif level, $G(3, 3)$. Motif-associated hubs at the deepest level of hierarchy were taken as key regulators.

An approach to find communities in vast networks is called the Louvain method. It maximises the modularity score for each community, where modularity measures how well nodes are allocated to communities. To do this, one must assess how much more closely connected nodes within a community are than they would be in a random network. The Louvain algorithm is a hierarchical clustering algorithm that performs modularity clustering on condensed graphs and recursively integrates communities into a single node [20]. The Louvain method for community detection is a method to extract communities from large networks created by Blondel et al. [21] The method is a greedy optimization method that appears to run in time $O(n \cdot \log n)$, where n is the number of nodes in the network. The modularity measure and a

hierarchical approach form the foundation of this method. Each vertex is initially given a community of its own. Every phase involves the local, greedy re-assignment of vertices to communities; in a random order, each vertex is transferred to the community with which it contributes the most to modularity. Each community is treated as a vertex on its own when there are no more vertices that can be assigned, and the procedure begins over with the combined communities. When there is only one vertex remaining or when the modularity cannot be enhanced further in a phase, the procedure comes to an end. Gamma, the resolution parameter, enables the discovery of communities at various resolutions. More and smaller communities are often produced by resolution parameters with higher values. Typically, lower values lead to fewer, bigger communities. When gamma (G) is set to 1, the original definition of modularity is returned. Keep take mind that the resolution parameter is used to determine the modularity value returned [21].

2.6. Functional analysis of annotation enrichment/over-representation

The functional biology relating the biological processes, molecular functions, cellular components and biological pathways against the framework of gene interaction networks can be understood more efficiently using annotation enrichment analysis [22]. The information available in multiple knowledge bases was used to find the over-represented annotations linked with genes/proteins (GO, Reactome, KEGG, etc.). Functional enrichment terms were taken from the STRING database to analyse the functionality of the genes. The gene's functional enrichment analysis allows us to comprehend all of the biological knowledge and functions that have been reported or predicted in the literature. STRING uses hypothetical testing, against a statistical background of specified genes list to perform automated pathway enrichment analysis. Gene annotations such as Gene Ontology (GO) uses hierarchically structured approach to classify genes into biological/cellular/molecular words (GO terms). Enrichments from KEGG pathways, Reactome pathways, UniProt words, Pfam, and InterPro domains data were also obtained in addition to the GO terms [10].

3. Results

3.1. AMR gene data acquisition and gene interaction network

A list of 235 unique AMR genes were extracted after filtering the redundant entries and AMR gene interaction data along with functional partners specific to *H. pylori* were retrieved from STRING database. Only 77 out of 235 genes had interaction data, thus we extended the network to acquire the greatest number of interactions between AMR genes and interacting functional partners (Table S1). At a medium confidence level, the final set of interaction data included 77 genes with 391 functional interactions (Fig. 1A).

3.2. AMR gene interaction networks exhibit scale-free topologies

The gene interaction network generated by STRING database consisted of 77 nodes and 391 edges. The topological properties of the genes in the interaction network were analysed using CytoscapeNetworkAnalyzer. Table S2 summarises the findings of each node's topological investigations, including k , $c(k)$, $C_B(k)$, $C_C(k)$, and $C_N(k)$. The number of edges is much more (p -value $\leq 10^{-6}$) than expected for a random network of the same size; the nodes were more connected than expected. The CytoNCA tool was used for measuring the Eigenvector centrality ($C_E(k)$). It was argued that the PPI network may be thought of as a small universe compared to a random graph, and that the proteins could be biologically important. The results of global topological metrics such as average degree (k), mean shortest path length ($mspl$), diameter (D), and average clustering coefficient, which are given in Table 1.

We found that six nodes with more than 20 direct interactors, followed by 29 nodes with 10–20 interactors, 24 nodes with 5–10

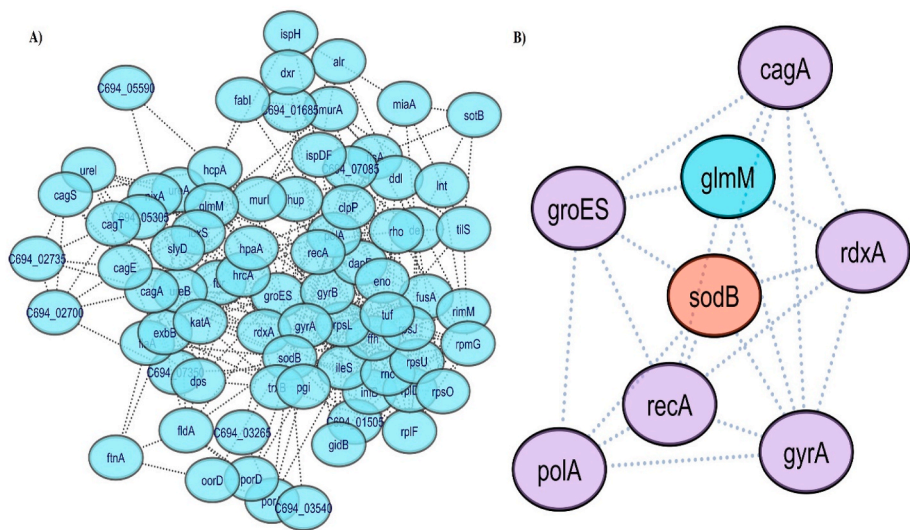


Fig. 1. (A) The interaction network of antibiotic resistant genes from *H. pylori* consisting with 77 nodes connected via 391 edges; (B) The backbone network constituting eight nodes connected via 21 edges, where *sodB* gene (red colour) is the backbone's most critical bottleneck, the genes *rdxA*, *gyrA*, *recA*, *groES*, *polA*, *cagA* (purple colour) are the first neighbours of the hub gene.

Table 1
The general network measurements for giant and backbone network networks.

Description	Giant Network	Backbone network
Number of nodes (N)	77	8
Number of edges (E)	391	21
Network Diameter (D)	5	2
Clustering coefficient	0.407	0.704
Network centralization	0.201	0.143
Shortest Paths	5826	56
Average Degree <k>	10.15	5.25
mean shortest path length (mspl)	2.323	1.25

Table 2
Top ten genes with the greatest number of functional interactors.

Sl. No	Gene	Degree	Functional neighbours
01	<i>sodB</i>	25	<i>flaA</i> , <i>dps</i> , <i>fldA</i> , <i>rdxA</i> , <i>trxA</i> , <i>kata</i> , <i>cagA</i> , <i>fur</i> , <i>luxS</i> , <i>groES</i> , <i>rns</i> , <i>gyrA</i> , <i>rpsJ</i> , <i>rplD</i> , <i>rplF</i> , <i>rpsO</i> , <i>gyrB</i> , <i>eno</i> , <i>polA</i> , <i>pgi</i> , <i>porA</i> , <i>ispDF</i> , <i>tuf</i> , <i>recA</i> , C694_03540
02	<i>gyrA</i>	25	<i>rplD</i> , <i>rpsJ</i> , <i>gyrB</i> , <i>rnc</i> , <i>rdxA</i> , <i>kata</i> , <i>cagA</i> , <i>rpsU</i> , <i>gidB</i> , <i>sodB</i> , <i>tuf</i> , <i>polA</i> , <i>recA</i> , <i>glmM</i> , <i>hup</i> , <i>oorD</i> , <i>porD</i> , <i>ileS</i> , C694_07085, <i>infB</i> , <i>ffh</i> , <i>rpsL</i> , <i>dopF</i> , <i>ftsA</i> , <i>fur</i>
03	<i>recA</i>	24	<i>rpsJ</i> , <i>rnc</i> , <i>kata</i> , <i>polA</i> , <i>sodB</i> , <i>tuf</i> , <i>glmM</i> , <i>ffh</i> , <i>rpsL</i> , <i>fusA</i> , <i>infB</i> , <i>ileS</i> , C694_07085, <i>groES</i> , <i>gyrA</i> , <i>eno</i> , <i>pgi</i> , <i>clpP</i> , <i>ddl</i> , <i>rho</i> , <i>dxr</i> , C694_01685
04	<i>groES</i>	23	<i>sodB</i> , <i>dps</i> , <i>cagA</i> , <i>trxB</i> , <i>nixA</i> , <i>rplD</i> , <i>rpsJ</i> , <i>slyD</i> , <i>hpaA</i> , <i>polA</i> , <i>hrcA</i> , <i>exbB</i> , <i>clpP</i> , <i>tur</i> , <i>recA</i> , <i>ureB</i> , <i>hup</i> , <i>glmM</i> , <i>ffh</i> , <i>ileS</i> , <i>ftsA</i> , <i>infB</i> , C694_03265
05	<i>infB</i>	22	<i>groES</i> , <i>rplF</i> , <i>pgi</i> , <i>gyrA</i> , <i>rpsO</i> , C694_01505, <i>rpmG</i> , <i>ileS</i> , <i>fusA</i> , <i>gidB</i> , <i>dapF</i> , <i>rpsL</i> , <i>ffh</i> , <i>rpsU</i> , <i>tuf</i> , <i>recA</i> , <i>polA</i> , <i>gyrB</i> , <i>rpsJ</i> , <i>rplD</i> , <i>rdxA</i> , <i>rnc</i>
06	<i>gyrB</i>	21	<i>rpsJ</i> , <i>rdxA</i> , <i>polA</i> , <i>sodB</i> , <i>gidB</i> , <i>tuf</i> , <i>recA</i> , <i>hup</i> , <i>dapF</i> , <i>ffh</i> , <i>rpsL</i> , <i>fusA</i> , <i>infB</i> , <i>ileS</i> , C694_07085, <i>gyrA</i> , C694_01505, <i>urea</i> , <i>pgi</i> , <i>ddl</i>
07	<i>polA</i>	20	<i>recA</i> , <i>sodB</i> , <i>gyrB</i> , <i>rpsJ</i> , <i>tuf</i> , <i>hub</i> , <i>ffh</i> , <i>ftsA</i> , <i>rpsL</i> , <i>fusA</i> , <i>infB</i> , <i>groES</i> , <i>fur</i> , <i>trxB</i> , <i>gyrA</i> , <i>miaA</i> , <i>ispDF</i> , <i>ddl</i> , <i>rho</i> , <i>fabI</i>
08	<i>tuf</i>	20	<i>ffh</i> , <i>sodB</i> , <i>recA</i> , <i>polA</i> , <i>gyrB</i> , <i>rpsJ</i> , <i>rplD</i> , <i>rpsL</i> , <i>fusA</i> , <i>infB</i> , <i>ileS</i> , <i>groES</i> , <i>rplF</i> , <i>trxB</i> , <i>gyrA</i> , <i>eno</i> , <i>clpP</i> , <i>rpmG</i> , <i>ddl</i>
09	<i>rpsJ</i>	20	<i>rplD</i> , <i>rnc</i> , <i>gyrB</i> , <i>polA</i> , <i>sodB</i> , <i>rpsU</i> , <i>recA</i> , <i>tuf</i> , <i>ffh</i> , <i>rpsL</i> , <i>fusA</i> , <i>infB</i> , <i>ileS</i> , <i>groES</i> , <i>rplF</i> , <i>gyrA</i> , <i>rpsO</i> , <i>rim</i> , <i>rpmG</i> , <i>rho</i>
10	<i>fusA</i>	20	<i>infD</i> , <i>rpsL</i> , <i>ileS</i> , <i>ffh</i> , <i>rpsU</i> , <i>tuf</i> , <i>recA</i> , <i>polA</i> , <i>gyrB</i> , <i>rpsJ</i> , <i>rplD</i> , <i>rnc</i> , <i>rplF</i> , <i>gyrA</i> , <i>rpsO</i> , <i>eno</i> , <i>clpP</i> , <i>rim</i> , <i>rpmG</i> , <i>ddl</i>

interactors, and 18 nodes with less than 5 direct interactors, out of a total of 77 nodes. Table 2 shows the top ten genes with the greatest number of functional interactors. The hub molecules in the network are the genes with the most direct contacts, and these genes are the governing points of molecular interactions. Hub genes can aid in the better understanding of molecular events and can be used as therapeutic targets for the development of new drugs. The top six genes with the highest degree value are *sodB* (Superoxide dismutase [Fe]), *gyrA* (DNA gyrase subunit A), *recA* (Protein RecA), *groES* (Co-chaperonin GroES), *infB* (Translation initiation factor IF-2) and *gyrB* (DNA gyrase subunit B). Table S3 shows that *sodB* was both hub (with the largest degree $k = 25$) and bottleneck (with the second highest $C_B(k) = 0.1191$) in the gene interaction network, while *groES* was solely a bottleneck (with the highest $C_B(k) = 0.11963$) but not a hub (with second highest degree). *SodB*, *groEL*, *gyrA*, *recA*, *polA*, *tuf*, *infB*, and *gyrB* genes with the top 10% biggest degree (k) or highest $C_B(k)$ were identified as hubs or bottlenecks in the core network (Table S2). They were very well connected to their network neighbours (average degree: 10.15) and had a great deal of influence over it (the average $C_B(k) = 0.0176$). The nodes (genes/proteins) with high BC should be the heavily used intersections of these nodes, and the links between them form a backbone network. Ten percent of the network's total number of nodes was chosen as the critical point of high BC [23]. A backbone network was built using these high BC nodes and the connections that connect them from the vast network. BC was initially defined as the network's nodes centrality. A network's high BC nodes are required to be traversed by the majority of shortest pathways. As a bottleneck in the network's nodes' communication with one another, these nodes perform. The backbone network consists of 8 high BC nodes, the size of which corresponds to their BC value and the 21 links between them (Fig. 3). Eight proteins, *sodB*, *groES*, *cagA*, *gyrA*, *recA*, *groEL*, *glmM* and *polA*, had the top 10% biggest degree (k) and highest $C_B(k)$. Without calculating the values of BC and CC, we can find out that *sodB* locates at the centre of the backbone network with the highest BC value and the largest degree. *SodB* has 6 neighbours: *groES*, *cagA*, *gyrA*, *recA*, *groEL* and *polA*. Table 3 lists the top 20 genes from the network analysis with the minimum average shortest path length and the highest $C_c(k)$ values. The data set revealed important topological characteristics including $C_B(k)$ and shortest path length. The relevance of a node's centrality in a network is indicated by its $C_B(k)$. The number of shortest paths from all other routes to other nodes that travel via this node, whereas shortest path length represents the shortest distance between any two interacting nodes in a network.

Table 3

List of top 20 genes from the network analysis with the minimum average shortest path length and the highest closeness centrality values.

Nodes	Avg shortest pathlength	C _c (k)
<i>sodB</i>	1.736842	0.575758
<i>groES</i>	1.763158	0.567164
<i>gyrA</i>	1.789474	0.558824
<i>recA</i>	1.789474	0.558824
<i>polA</i>	1.868421	0.535211
<i>tuf</i>	1.881579	0.531469
<i>infB</i>	1.894737	0.527778
<i>gyrB</i>	1.894737	0.527778
<i>rpsJ</i>	1.907895	0.524138
<i>cagA</i>	1.907895	0.524138
<i>rdxA</i>	1.947368	0.513514
<i>rpsL</i>	1.960526	0.510067
<i>glmM</i>	1.960526	0.510067
<i>fusA</i>	2	0.5
<i>ffh</i>	2	0.5
<i>katA</i>	2	0.5
<i>rplD</i>	2.013158	0.496732
<i>ileS</i>	2.039474	0.490323
<i>rnc</i>	2.052632	0.487179
<i>ureB</i>	2.078947	0.481013

To comprehend the network’s signal propagation and structural features, topological properties were examined. The probability of degree distribution $p(k)$, clustering coefficient $c(k)$, and neighbourhood connectivity distribution $C_N(k)$ as a function of degree were found to demonstrate power law or fractal structure, indicating that the network topology is hierarchical and scale free. To fit the power law on the data sets concerned with distinct topological parameters in the investigation, a typical statistical fitting approach was used (Fig. S1).

3.3. MCODE-derived gene complexes

Genes that act as drug targets can be believed of as vents that allow for network variations. Though there are many drug-actionable genes for AMR, we only focus on those that can be traced in dense clusters using the MCODE algorithm (keeping in mind the assortative nature of the AMR) with strict parameters (haircut = TRUE; degree cutoff = 2; node score cutoff = 0.2; K-Core = 2; maximum depth = 100). After imperilling the main network to MCODE, six such clusters or gene complexes were discovered (Table 4). Among the six clusters, the cluster C1 has shown dense interactions with 15 genes and 78 functional interactions (Fig. 2). Whereas the clusters C2, C4 and C5 have shown 6, 5 and 9 with 12, 7 and 19 edges, respectively.

3.4. Key regulators in the AMR network

The hubs that serve as backbone of the network were considered as KRs which are deeply rooted and contribute to networks local and global stability. They play an important role in preserving the network’s functional and structural integrity, as well as defending it from threats. Apart from signal transmission and reception, KRs also act as a medium

Table 4

Clustering of AMR genes in *Helicobacter pylori* and their functional relationships.

Clusters	MCODE score	No of nodes	No. of edges	Functional partners
C1	11.143	15	78	<i>polA</i> , <i>recA</i> , <i>rnc</i> , <i>rpsU</i> , <i>infB</i> , <i>rpmG</i> , <i>ileS</i> , <i>gyrB</i> , <i>tuf</i> , <i>rpsJ</i> , <i>rplE</i> , <i>rpsO</i> , <i>ffh</i> , <i>fusA</i> , <i>rplD</i>
C2	4.8	6	12	<i>C694_02735</i> , <i>murI</i> , <i>cagT</i> , <i>cagS</i> , <i>cagE</i> , <i>C694_02700</i>
C4	3.5	5	7	<i>sodB</i> , <i>rdxA</i> , <i>rpsL</i> , <i>gyrA</i> , <i>dapF</i>
C5	3.25	9	13	<i>murA</i> , <i>ureA</i> , <i>nixA</i> , <i>glmM</i> , <i>exbB</i> , <i>ureI</i> , <i>luxS</i> , <i>fabI</i> , <i>cagA</i>

in cross-talks between nodes, even when they are separated by a large distance. Hubs, which have high centrality values, play a key role in directing the flow of signal/information in our modular, hierarchical, scale-free AMR gene interaction network. Changing popularity of hubs with various activities and regulating systems, on the other hand, indicates that all prominent hubs cannot function as KRs.

In order to locate the KRs in the network, we used the Louvain technique of modularity (Q) maximisation for community detection algorithm to trace down hubs of degree 20, which were traced down in modules at every level of hierarchy up to the level of motifs G (3, 3). All of these communities exist at various hierarchical levels, and their modularity values result in motifs (Fig. 3).

3.5. Gene-ontology-based overrepresentation analysis

All 77 genes involved in the network were subjected to enrichment analysis in ShinyGO and STRING databases according to GO-BP (biological processes), GO-MF (molecular functions), GO-CC (cellular components) and KEGG pathways (Fig. 4). BPs are GO terms that represent the capability of many cellular interactions and are vital for the survival of the cell [4,24]. The molecular functions of the gene product, such as binding and catalysis, are known as MFs. CCs are cell structures and biomolecular complexes [25].

The most abundant GO-BP terms were concerned with biological regulation (cellular component organization or biogenesis (GO:0044085), gene expression (GO:0010467), cellular process (GO:0009987), etc.), metabolic processes (response to reactive oxygen species (GO:0000302), urea metabolic process (GO:0019627), nitrogen cycle metabolic process (GO:0071941), reactive oxygen species metabolic process (GO:0072593), nitrogen compound metabolic process (GO:0006807). etc) and cellular process (cellular macromolecule biosynthetic process (GO:0034645), cellular nitrogen compound metabolic process (GO:0034641), cellular biosynthetic process (GO:0044249), etc. The most enriched GO-CC categories were extra-cellular region (GO:0005576), intracellular organelle (GO:0043229), non-membrane-bounded organelle (GO:0043228), cellular anatomical entity (GO:0110165), etc. While most enriched GO-MF were racemase and epimerase activity (GO:0016854), acting on amino acids and derivatives (GO:0016855), amino-acid racemase activity (GO:0047661), racemase and epimerase activity (GO:0016854), antioxidant activity (GO:0016209), isomerase activity (GO:0016853), nucleoside-triphosphatase activity (GO:0017111), DNA binding (GO:0003677), pyrophosphatase activity (GO:0016462), hydrolase activity, acting on acid anhydrides (GO:0016787), hydrolase activity, nucleic acid binding (GO:0003676), metal ion binding (GO:0046872), cation binding (GO:0043169), ion binding (GO:0043167), catalytic activity (GO:0003824), organic cyclic compound binding (GO:0097159) and heterocyclic compound binding (GO:1901363). Epithelial cell signaling in *H. pylori* infection (hsa05120) the over-represented KEGG molecular pathways besides pathways involved in AMR.

4. Discussion

To explore the genes that contribute in AMR of *H. pylori* a gene interaction network was constructed and analysed. The MDR mechanism of *Enterococcus faecalis* V583 was elucidated using gene interaction network analysis and showed that *mraY*, *pbpC*, *murE*, *murC* and *murD* possessed the highest interactions (node degrees) which are considered as hub genes that could be exploited as potential drug targets [26]. Gene/protein interaction network studies were employed deliberately to decipher mechanisms involved in the pathogenesis and drug resistance in bacterial species [27]. Studies by Miryala et al. [4] have demonstrated the molecular-level understanding of interactions in the CVDs between the human and the *E. coli* genes and found that important genes with vital interactions can be considered as hub molecules and can be exploited for new drug discovery. Gene interaction network studies have

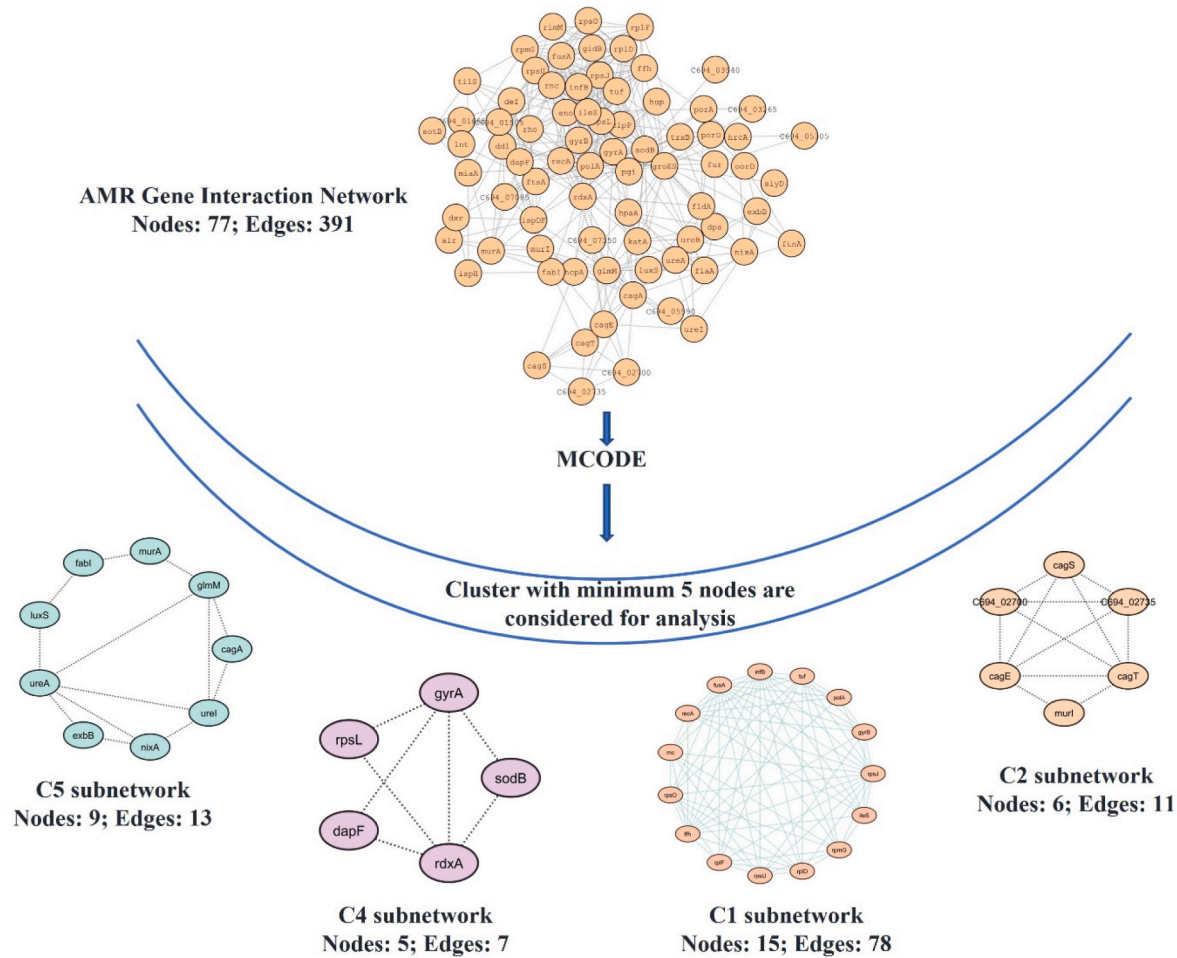


Fig. 2. MCODE Gene clusters (C1, C2, C4 and C5) obtained from the MCODE algorithm. Only those complexes with a minimum of five nodes are considered for this process.

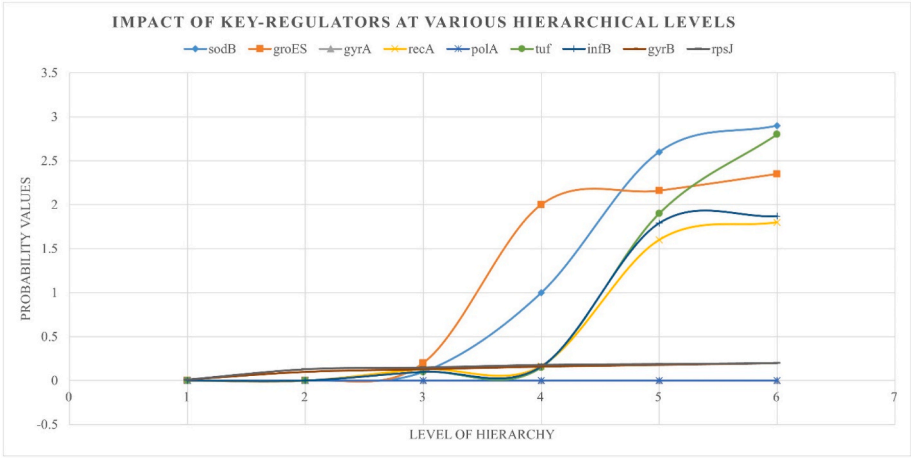


Fig. 3. The Louvain method of modularity (Q) maximisation is used to detect communities. As we proceed deeper into the network's hierarchy, a probability graph will highlight the increasing impact of key regulators.

provided profound insights into the MDR mechanism in *Salmonella enterica* serovar Typhi CT18 [28]. In this crucial situation, bio-informaticians and computational biologists are effectively advancing clinical research and medicinal development [29]. Topological analysis was used to investigate the potential critical genes. The essential parameters for evaluating the nodes in the gene interaction network were

degree (k) and $C_B(k)$, two well-known network measurements [23]. Initially, our core interaction network had 77 proteins. We chose eight proteins for the backbone network based on essential topological parameters (k and $C_B(k)$) in a large network: eight with big degrees, seven with high $C_B(k)$, and six with both large k and high $C_B(k)$. As the probability of degree distribution $p(k)$, clustering coefficient $c(k)$, and

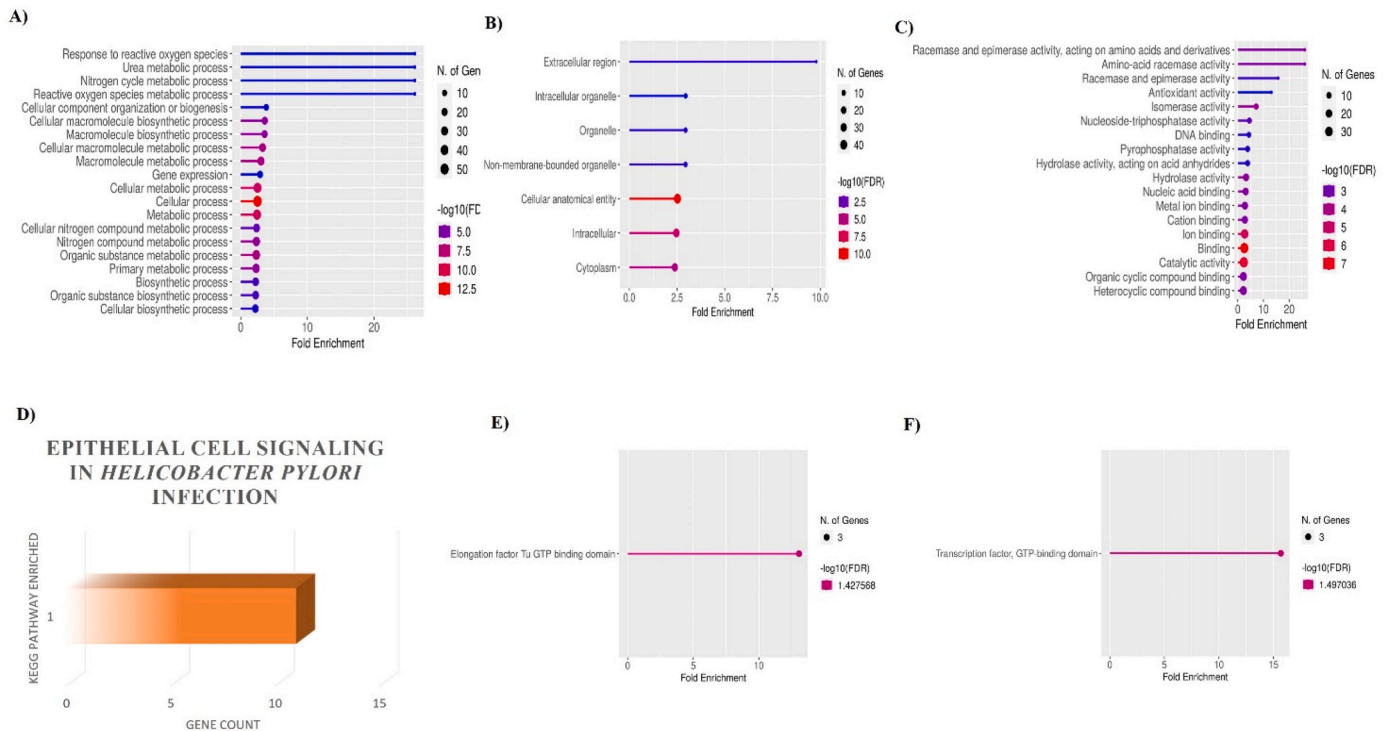


Fig. 4. Functional enrichment analysis in ShinyGO and STRING databases according to (A) GO-BP (biological processes), (B) GO-MF (molecular functions), (C) GO-CC (cellular components), (D) KEGG pathways, (E) Pfam domain and (F) InterPro domain.

neighbourhood connectivity distribution $C_N(k)$, all of which have a fractal nature as a function of degree, our investigation into the interactions and topology of AMR gene interaction network also revealed its hierarchical, scale-free, and assortative nature. The network's centrality measures, such as $C_B(k)$, $C_C(k)$, and $C_E(k)$, were likewise fractal as a function of degree, with positive exponents indicating the importance of strongly linked nodes (hubs) in regulation of network. The topological feature of the network with great attention is betweenness, which measures the total number of nonredundant shortest paths going through a certain node or edge. Betweenness was originally introduced to measure the centrality of the nodes in networks [30]. By definition, most of the shortest paths in a network go through the nodes with high betweenness. Therefore, these nodes become the central points controlling the communication among other nodes in the network. Another study revisited the original meaning of betweenness as a measure of the centrality of the nodes in a network. It was hypothesized the protein networks (in particular, regulatory networks) in analogy to a transportation network, proteins with high betweenness are similar to heavily used intersections, such as those leading to major highways or bridges. If these major intersections were blocked, there would be huge traffic jams, causing the whole transportation system to fail. Therefore, we called these high-betweenness proteins bottlenecks, and hypothesized that these bottlenecks, just like hubs, represent important points in biological networks as well. For simplicity, we defined protein bottlenecks as the proteins with the highest betweenness; hubs, as the proteins with the highest degree [31]. Therefore, betweenness is one of the most important topological properties of a network. It measures the number of shortest paths going through a certain node. Therefore, nodes with the highest betweenness control most of the information flow in the network, representing the critical points of the network.

The repeating basic geometrical units embodying their evolutionary principles—"motifs"—are "patterns for which the probability P of appearing in a randomised network an equal or greater number of times than in the real network is lower than a cut-off value" in complex networks [32]. Given the vastness and complexity of our AMR gene

interaction network, it is critical to examine the implicated motifs and understand their impact on system-wide dynamics [16]. Using the Louvain method of modularity (Q) maximisation algorithm for community detection, we traced a total of nine high-degree hubs (degree ≥ 20) up to the level of motifs $G(3, 3)$, as this clustering algorithm is more suitable for large complex networks than other common community detection algorithms. The network was divided into 7 layers, with average modularity values (Q) decreasing from the first to the seventh level with hubs present in motifs at the deepest levels serving as major regulators (*sodB*, *gyoES*, *gyrA*, *recA*, *polA*, *tuf*, *infB*, *gyrB* and *rpsJ*). As we progressed deeper into the network, the regulating influence of KRs developed stronger.

***sodB* as critical hub of the gene interaction network:** *H. pylori* which is exposed to hostile environmental conditions (acidic pH and active immune responses) colonizes the mucus layer overlapping to gastric epithelial cells [33]. In *H. pylori*, oxidative stress is induced by the host's strong immunological response. Furthermore, during bacterial respiration and metabolism, reactive oxygen species such as, superoxides are produced. It's no surprise that enzymes involved in the detoxification of reactive oxygen species are significant *H. pylori* colonisation factors, given this organism's microaerophilic needs and limited oxygen tolerance. Only one superoxide dismutase (SOD), the iron-cofactored *sodB* protein, is expressed by *H. pylori*. *SodB* expression is required for *H. pylori* colonisation in the stomach and proliferation under microaerophilic conditions [34]. The *sodB* of *H. pylori* is associated with the cell surface, consistent with a role in protection against exogenously produced superoxide that incurred during an oxidative burst via a host inflammatory response [35].

Iron is required by practically every living organism, including *H. pylori*, for a variety of metabolic pathways such as electron transport, DNA replication, and amino acid synthesis [36]. *H. pylori* has access to host dietary iron within the gastric niche, but it must compete with the host for this scarce resource [37]. *H. pylori* can use haemoglobin, transferrin, heme, and lactoferrin as alternative sources of nutritional iron when iron is scarce [38]. As a result, *H. pylori* responds to iron

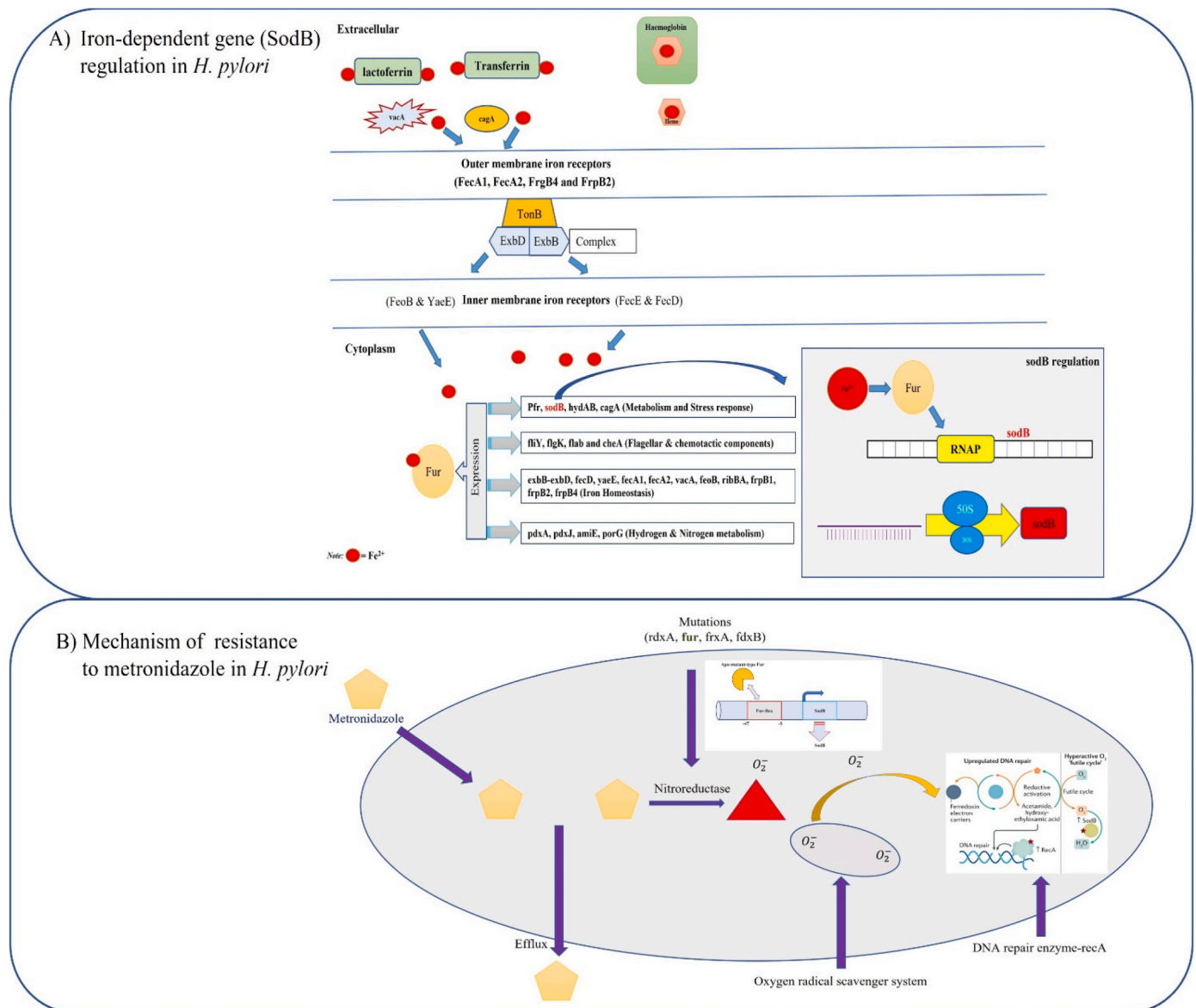


Fig. 5. (A) Schematic diagram illustrating the iron-dependent gene (*sodB*) regulation in *H. pylori*; (B) mechanism of metronidazole resistance acquired in *H. pylori* due to overexpression of *sodB* upon binding of Ferric Uptake Regulator (Fur) regulated by fur gene.

deficiency by upregulating a slew of biochemical pathways (Fig. 5A) dedicated to various iron acquisition tasks; thus, *H. pylori*'s ability to detect and respond to environmental iron concentrations is important to its survival [39]. SOD, which catalyse the conversion of superoxide species into oxygen and hydrogen peroxide, are another mechanism for dealing with oxidative stress caused by elevated iron levels [40]. Fur transcriptionally regulates the SOD found in *H. pylori*, *sodB*, such that apo-Fur binds to the promoter region of *sodB*, preventing RNA polymerase binding (Fig. 5A). *H. pylori* can appropriately respond to high iron levels with a set of proteins designed to reduce iron toxicity as a result of apo-Fur control of both *sodB* and fur. Apo-Fur has been demonstrated to suppress *hydABCDE*, an operon that encodes a Ni/Fe hydrogenase, implying that apo-Fur is also involved in hydrogen metabolism regulation in *H. pylori* [41]. Furthermore, apo-Fur-dependent repression of genes (including *sodB* and those encoded in the *hydABCDE* operon) is thought to have evolved because their protein products use iron as a cofactor, and apo-Fur-dependent repression ensures that these proteins are produced only when sufficient levels of that cofactor are available [40].

The host's immunological response includes reactive oxygen species

induced damage against pathogenic microorganisms. *H. pylori* infection causes a high oxidative stress response in the host, according to many studies [41]. *H. pylori* relies on a diverse set of detoxifying enzymes, including *sodB*, catalase (KatA), and neutrophil-activating protein (NapA), to withstand the effects of the host's production of reactive oxygen species [42]. However, molecular oxygen occurs in the *H. pylori* internal micro-aerophilic environment, which is not totally anaerobic, and competes with metronidazole (Mtz) for electrons, leading in "futile cycling" – reoxidation of Mtz radicals and generation of superoxides, which causes DNA damage [43]. Scavenging these superoxides is an obvious Mtz resistance strategy, and resistant strains appear to take advantage of this possibility. Increased activity of the oxygen radical scavenger system, enhanced activity of DNA repair enzymes, reduced absorption and increased efflux, and reduced activity of the nitroreductases have all been reported as plausible pathways for Mtz resistance in *H. pylori* [44]. Overexpression of *sodB* mediated by mutant-type Fur has been shown to fortify *H. pylori*'s Mtz resistance that is not dependent on NADPH nitroreductases (RdxA). Singh et al. [45], has demonstrated that along with other genes, *sodB* is also significantly upregulated and specifically induced during infection to the host. Earlier

reports suggest that *sodB* assist *H. pylori* to combat oxidative stress and ensures its adaptation to survive within the host gastric environment [1]. Tsugawa et al. [42], demonstrated the structural basis of Fur based on homology modelling that comprises three functional domains: DNA-binding domain near N-terminal, iron-cofactor binding domain (HHDHXCXC) and the oligomerization domain at C-terminal. Their studies revealed mutations in DNA-binding domains (C78Y) and oligomerization domain (P114S & N118H), and predicted that mutations may affect the affinity of the Fur protein of the Fur-Box supporting our hypothesis of antimicrobial resistance. Earlier studies have reported that, in addition to the hydroxyl radical damage, superoxide toxicity may include the reduction or oxidation of other compounds and interaction with nitric oxide (NO) to form peroxynitrite. Because NO level in gastric juice is fairly high, superoxide-dependent peroxynitrite toxicity may be an important factor in the defective cocolonization abilities of *sodB* mutants [34]. Other proteins such as pyruvate oxidoreductase, nicotinamide adenine dinucleotide dinucleotide phosphate (reduced form) flavin oxidoreductase (FrxA), and ferredoxin-like protein (FdxB) have recently been discovered to be involved in the activation of Mtz. The reports available have shown a link between the inactivation of these proteins and Mtz resistance [46,47]. On the other hand, the RdxA independent mechanisms may play only a limited part in Mtz resistance or that they are only involved in the transition to high-level resistance [42,48]. Furthermore, Fe-SOD overexpression has been linked to Mtz resistance in Mtz-resistant strains of the protozoan parasite *Entamoeba histolytica* the causative agent of human amoebiasis [49]. Overexpression of *sod* may influence the Mtz resistance mechanism in many bacterial species, according to these results. FecA1 deletion results in increased Mtz sensitivity in Mtz-resistant strains (KS0048 and KS0145), suggesting that FecA1 is also involved in the development of Mtz resistance. Here we propose the hypothesis that aberrant increase of *sodB* expression resulting from mutations of Fur would be a novel mechanism of Mtz resistance of *H. pylori*. The upregulation of *sodB* by reduced binding of mutant Fur or single base-pair substitution in *sodB* promoter region leads to hyperactivity of oxygen 'futile cycle' that leads in regeneration of inactive drug compounds and protection against oxidative reactions (Fig. 5B).

5. Conclusion

Drug resistance in *H. pylori* has been studied less thoroughly, which could indicate more complex physiological alterations that are still unknown. The gene interaction network analysis of AMR genes in *H. pylori* to explore drug resistance patterns, due to their substantial influence over antibiotic resistance, eight such hubs (*sodB*, *groEL*, *gyrA*, *recA*, *polA*, *tuf*, *infB*, and *gyrB*) exist at the deepest level of hierarchies and are considered KRs. As these hub nodes have more direct interactions and could be more important in evaluating the molecular level interactions of the AMR genes. We demonstrated how the *sodB*, which are involved in the response to reactive oxygen species, urea metabolic process, nitrogen cycle metabolic process, reactive oxygen species metabolic process, regulated by Fur gene/proteins, claim a major authority over regulation and signal propagation in the AMR, as well as over interacting hubs (network stability), using a comprehensive methodology and review of the literature. The most studied antibiotic resistance mechanisms are related to chromosomal mutations that disrupt antibiotic activity via target-mediated mechanisms. Other biological factors that contribute to *H. pylori* drug resistance have received less attention, which could indicate more complex physiological changes needs to be explored. Overall, we believe the findings given in this work will serve as an excellent starting point and stimulate *in vitro* and *in vivo* studies to develop new treatment options for *H. pylori* infections.

Authorship contributions

P. G. supervised and planned the experiments, performed the

experiments and wrote the paper draft. P. G., and T. S. G. carried out the experiments and contributed to the interpretation of the results. S. K. H. S. and B.K.K. contributed in manuscript editing. P. G. took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis and manuscript.

Data availability

All the data we generated in this paper is available in the body of the manuscript, supporting tables and supplementary material. We do not have any ethical or legal considerations making or not to make our data publicly available.

CRediT authorship contribution statement

Pavan Gollapalli: Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **H.S. Santoshkumar:** Writing – review & editing, Formal analysis, Data curation. **Krishna Kumar Ballamoole:** Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgments

The authors are thankful to Nitte (Deemed to be University), Mangalore, India for providing all the facilities to complete this work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2022.105737>.

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